# Graphical Abstract

## An Explainable and Robustness-Aware Convolutional Neural Network Framework for Three-Class Face-Mask-Compliance Classification from Public Surveillance Imagery

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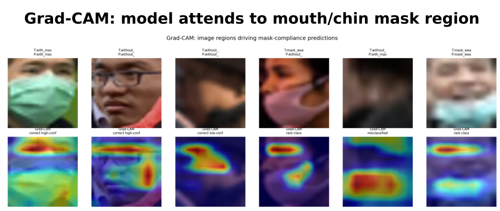


Three-Class Face-Mask Compliance

Pipeline
1. Kaggle FMD: 853 images, PASCAL-VOC
2. Crop faces -> 4,072 crops, 3 classes
3. Softened class weights (5.1/1.0/2.1)
4. Custom CNN (259k params, 64x64)
5. Grad-CAM + SHAP + robustness

Key results
- Test accuracy: 92.5%
- with_mask F1 0.96, without_mask F1 0.88
- Rare incorrect-mask class hard (F1 0.32)
- Explanations focus on lower face
- Robust to blur; sensitive to heavy noise

# Highlights

**An Explainable and Robustness-Aware Convolutional Neural Network Framework for Three-Class Face-Mask-Compliance Classification from Public Surveillance Imagery**

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- Three-class mask-compliance classification separates the high-risk incorrectly-worn case from mask and no-mask.
- A from-scratch custom CNN reaches 92.46% test accuracy on 4,072 cropped faces.
- Softened class weighting prevents minority-class collapse caused by severe imbalance.
- Grad-CAM and SHAP confirm the model attends to the mouth and chin mask region.
- Robustness testing reveals graceful blur tolerance but strong sensitivity to additive noise.

# An Explainable and Robustness-Aware Convolutional Neural Network Framework for Three-Class Face-Mask-Compliance Classification from Public Surveillance Imagery

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**Abstract**

Face mask wear was identified as a critical, non-pharmaceutical remedy in the field of respiratory disease and computer vision-aware monitoring of mask compliance in public settings is an increasingly applicable problem.

But most published face-mask studies treat the problem as a binary mask versus no-mask classification while relatively few actually systematically assess the more difficult three-class setting that recognises a mask not worn correctly, which is in fact precisely the condition that undercuts the protective purpose of masking.

This study tackles a 3 class problem, namely, identifying masked individuals who are correctly worn (for example, the non-mask, cropped persons), incorrectly worn (one that has not been properly worn), and absent (those who have not been fitted with the mask as intended); it combines predictive assessment with explainability and robustness analysis rather than just reporting the accuracy data.

A custom convolutional neural network is trained with 4,072 face crops generated from the 853-image PASCAL-VOC-annotated Kaggle Face Mask Detection data via softened class weighting to alleviate extreme class imbalance.

With a matched stratified test set, the model returns a total accuracy of 92.46%, one weighted F1-score of 0.93, two well-represented classes ROC-AUC of 0.982 and 0.993 per-class. Grad-CAM and SHAP analyses confirm that predictions depend on the mouth and chin region vs background artefacts and robustness testing shows gentle degradation at blurring and occlusion but high sensitivity to heavy additive noise.

The key contribution is the reproducible, explainable, and robustness aware baseline, which highlights that the exceptionality of the rare incorrectly-worn mask class is the greatest open challenge for deployment.

*Keywords:* Face mask detection, Convolutional neural networks, Transfer learning, Explainable AI, Image classification, Class imbalance

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# 1. Introduction

## *1.1. Application Background and Practical Importance*

Face masks were among the most popular non-pharmaceutical interventions for the recent respiratory-disease outbreaks, and public health authorities continued to rely on wearing them to reduce airborne transmission in crowded indoor spaces. In practice, the masks' effectiveness does not just depend on whether the mask is on, but whether it is worn correctly on both the nose and the mouth, because a mask worn incorrectly gives a much less protective effect and at the same time gives the appearance of safety. In high-throughput environments, e.g., airports, hospitals, public transport hubs, and large retail premises where compliance is monitored closely in visual form, manual visual checking for compliance of a mask is labor-intensive and inconsistent between human observers, which cannot be achieved at the scale needed for continuous monitoring. The people and organisations most impacted by this issue are safety-oriented facility operators in charge of ensuring that workers are safe and sound, frontline staff who are responsible for enforcing policies and the wider society as a whole where collective risk increases when compliance is poor. Delayed or erroneous evaluation of mask compliance can have a tangible impact, as not noticing non-compliance would lead to localized transmission clusters which may be expensive to address. Traditional manual methodologies also put enforcement officials face-to-face and repeatedly with people that may be non-compliant which exacerbates their threat. Automated image analysis presents an appealing alternative — the system can run continuously, make consistent decisions based on agreed criteria, and only call out the cases that need checking by humans. A good automated system can thus serve as a decision-support and prioritisation tool and not as a fully autonomous enforcement mechanism, the one that has been the frame adopted throughout the study.

### 1.2. Machine Learning and Computer Vision Context

Image-based machine learning has grown so quickly over the last 10 years that convolutional neural networks have become the general paradigm for visual recognition problems. The key benefit of a convolutional neural network is that it learns hierarchical visual features straight from pixels; this eliminates the use of hand-crafted feature extraction methods commonly found in previous computer-vision pipelines. Early convolutional layers usually model low-level features like edges, colours, and textures, while deeper layers combine elements of all of these structures into higher level patterns that can be related to object parts and whole objects. In the face-mask task, this hierarchical ability is particularly useful, seeing as the difference between wearing good masks, badly, or exposing faces is encoded in local texture, colour, and the spatial organisation of the nose, mouth. In addition, convolutional networks are spatially equivariant with only minor changes of the position of the respective facial region in the cropped image. However, being data-hungry, the same flexibility which gives deep networks power leads them to overfit; training a deep model from random initialisation typically necessitates lots of labelled datasets in order to prevent overfitting. This limitation becomes particularly evident for the face-mask domain, where publicly available annotated datasets are small and fairly imbalanced across the three compliance classes. Accordingly, the methodological problems tackled by this study concern the ability to achieve consistent convolutional capabilities under genuine limitations of small and imbalanced datasets.

### 1.3. Transfer Learning for the Selected Problem

When a small number of images are labelled, it is hard to train a CNN from random initialisation as the model has many parameters and even less information from which to estimate them. Transfer learning solves this problem by reusing a model which trained on a very large general image set, e.g., ImageNet dataset. The pre-trained backbone is capable of producing general-purpose visual features - edge, texture and shape detectors - that are useful across a large number of downstream domains, such as facial analysis. Two common approaches are feature extraction (which consists of freezing the backbone and training only a new classification head), and fine-tuning (which also includes updating some backbone layers on the target data). In this regard, lightweight methods like MobileNetV2 are ideal for compliance monitoring, since the inverted-residual architecture gives competitive accuracy and fewer parameters, which is important for edge deployments.

Deeper residual architectures like ResNet50 provide higher representational capacity by skip connections, which makes training in deep stacks easier but requires a higher computational load. The current study places a customized convolutional baseline beside these transfer-learning alternatives, such that trade-off between accuracy and efficiency can be investigated explicitly rather than taken for granted. Following the protocol of keeping motivation as the main focus in the Introduction, a more detailed technical description for each architecture is deferred to the Methodology.

### *1.4. Explainable AI and Trustworthiness*

Predictive accuracy alone is not enough to gain trust for a model that could help guide safety-relevant decisions in public spaces. A model can achieve high headline accuracy and yet rely on spurious shortcuts – such as background colour or image-acquisition artefacts – that just happen to correlate with the label in training data but do not generalise. Explainable artificial intelligence methods can then expose this vulnerability by revealing which image regions most influence a prediction, allowing a human reviewer to judge whether the model's reasoning is plausible. Grad-CAM is a popular gradient-based technique that produces class-specific heatmaps highlighting the spatial regions that drive a particular output. SHAP offers a complementary, theoretically grounded attribution of a prediction to individual input features or regions: it separates evidence that supports a class from evidence that opposes it. In the case of the face-mask problem it is clear what kind of behaviour is ideal, namely that the model should attend to the mouth, nose, and chin regions where mask presence and fit are visible, rather than to the background or hair. As incorrect or unexplained predictions in a surveillance or safety context may result in unfair treatment of individuals or misplaced confidence in compliance levels, interpretability is treated as a first-class evaluation goal in this study, rather than an optional extra. Specifically, this work uses Grad-CAM and SHAP and assesses whether the regions highlighted are consistent with domain expectations.

### *1.5. Research Gap*

Following a systematic reading of the face-mask literature, there were several recurrent limitations that inform and facilitate the work presented here. The first thing is that the majority of existing studies have treated the task as binary classification: mask (or no-mask) and never considered separately the incorrectly-worn-mask class, which is the most safety-relevant and visually subtle class to consider. Second, although class imbalance exists in the entirety of public mask datasets, most works do not employ a class imbalance-handling method or mention it explicitly, so they cannot be trusted with minority class outcomes. Third, explainability is often restricted to a relatively few Grad-CAM heatmaps and quantitative or dual-method explanation with both Grad-CAM and SHAP is infrequent in this space. Fourth, hardly any studies assess robustness to realistic degradation of images, such as blur, noise, brightness change and occlusion, despite data showing that real surveillance images are routinely degraded. Fifth, statistical reliability, external validation, and explicit efficiency analysis are reported inconsistently, and thus, arguments for practical deployability are undermined. Not all of these gaps are present in each paper consistently, and it would be inaccurate to claim that no paper touches the individual gaps but it is rare that they intersect in a single, comprehensive manner in a single evaluation. The current study addresses this combined gap by solving the three-class problem in an explicit manner using imbalance handling, dual explainability and systematic robustness analysis in a single reproducible pipeline.

### *1.6. Problem Statement*

Using a comparative and explainable convolutional-neural-network framework, we tackle the automatic classification of the mask-compliance state of a human face from the cropped facial image into one of three classes—correctly worn mask, incorrectly worn mask, and no mask. The input includes a colour image of a single face cropped from a public surveillance-style photograph and the prediction target is the discrete compliance class of that face. The model has three classes, and the biggest challenge faced by models has been the extreme imbalance between many correctly worn-mask examples versus very few incorrectly worn-mask examples. The study looks at whether a carefully regularised custom convolutional network can yield good performance on both most frequent classes while giving a genuine account of how hard it is for a rare class. The intended practical use of the resulting system is not to be something like an autonomous enforcement mechanism but rather an automated screening and prioritisation aid that flags likely non-compliance for subsequent human verification..

### 1.7. Aim and Objectives

This study focuses on creating and testing an interpretable model with robustness-aware convolutional-neural-network mechanism for the predicted three-class classification of facial face-mask compliance obtained from cropped facial images. The precise goals are

- To analyze and preprocess the chosen public face-mask dataset, which entails conversion of object-detection annotations to a classification dataset of cropped faces.
- To develop a custom convolutional neural network to be used as a replicable baseline for 3-class mask-compliance classification.
- To apply an explicit class-imbalance treatment and report its impact on minority-class behaviour.
- To measure predictive performance by accuracy, per-class precision, recall, F1-score and ROC-AUC, focusing on the minority class.
- To analyze training behaviour, convergence, and computational characteristics of the model.
- To explain predictions using both Grad-CAM and SHAP and to assess whether the explanations are domain-meaningful. To evaluate robustness under blur, noise, brightness change, and occlusion, and to discuss the resulting deployment implications

### 1.8. Research Questions

The study is organised around the following five research questions, which are carried through the Results and Discussion sections.

- RQ1: How does the conversion of the detection-style dataset into cropped face images, together with augmentation and class-imbalance handling, affect the composition and learnability of the three-class dataset?
- RQ2: What overall predictive performance can a custom convolutional neural network achieve on the three-class mask-compliance task under realistic data constraints?

- RQ3: How does performance vary across the three classes, and what does the confusion structure reveal about the model's reliability on the safety-critical incorrectly-worn-mask class?

- RQ4: Do Grad-CAM and SHAP explanations indicate that the model bases its decisions on the mouth and chin mask region rather than on irrelevant background features?

- RQ5: How robust is the model to common image degradations, and what does its efficiency profile imply for practical deployment?

### *1.9. Contributions and Novelty*

This study makes a set of focused contributions that together address the combined gap identified above. The contributions are not the mere use of standard convolutional networks, but rather the rigorous and honest evaluation of a three-class compliance model under conditions that mirror real deployment constraints..

- A fully reproducible pipeline that transforms the PASCAL-VOC-annotated Kaggle Face Mask Detection dataset into a 4,072-image three-class classification dataset through bounding-box face cropping.

- A custom convolutional baseline trained with softened class weighting, with an explicit demonstration that naive balanced weighting causes minority-class collapse whereas softened weighting does not.

- A dual-method explainability analysis using both Grad-CAM and SHAP, assessed against the domain expectation that attention should fall on the mouth and chin region.

- A systematic robustness evaluation across blur, additive noise, brightness change, and occlusion at multiple severity levels.

- A transparent account of the persistent difficulty of the rare incorrectly-worn-mask class, framed as the principal open challenge rather than concealed behind aggregate metrics.

### 1.10. Report Organization

The rest of this report is structured as follows. Section 2 reviews the related literature across image-based machine learning in the domain, convolutional and transfer-learning methods, preprocessing and augmentation, evaluation practice, explainability, and robustness, and it presents a comparative literature matrix. Section 3 describes the complete methodology, including the dataset, data preparation, model development, training procedure, evaluation framework, explainability methodology, and robustness analysis. Section 4 reports the experimental results organised by research question, and Section 5 interprets those results, discusses limitations, and proposes future work. Section 6 concludes the report.

## 2. Literature Review

### 2.1. Image-Based Machine Learning in the Application Domain

Face-mask and facial-condition recognition image analysis has evolved through three large steps, which resemble the greater history of computer vision. At the early stage, traditional image-processing methods depended on hand-crafted descriptors including colour histograms, Haar-like features, and edge maps for face discovery as well as coverage inference, and, in terms of pose variation, illumination, and background variation, they were brittle. The second phase involved the classical machine-learning classifiers, specifically the support vector machines and decision ensembles, that were trained with engineered features, which improved accuracy but still relied directly on the well-crafted manual feature extraction. The third phase is now dominated by deep convolutional architectures that learn features autonomously, which have greatly improved the level of performance achievable for mask-related tasks. On practical terms, the deep-learning phase benefits from a single trained network with complete pipeline from raw pixels to class label without the need of separating feature engineering stages. However, the ongoing challenge is that deep models thrive on big and balanced datasets; public mask datasets rarely satisfy this criterion. The tradeoff between the strength of deep learning and the lack of well-balanced domain data drives the imbalance-aware and explainability-aware design selected in the current study.

### 2.2. CNN-Based Methods for the Selected Task

Many studies utilize convolutional networks to specifically target the problem of mask detection and classification. Nagrath et al. based a real-time mask detection system using a single-shot detector and MobileNetV2 classifier, indicating that lightweight backbones can support practical throughput. Loey et al. combined a deep transfer-learning feature extractor and classical machine-learning classifiers, indicating that hybrid pipelines can work well when training data is limited. Sethi et al. concluded single-stage detection was particularly effective and achieved strong binary accuracy but, like many studies, focused solely on the mask vs. no-mask dichotomy, rather than in the three-class context. Chowdary et al. fine-tuned InceptionV3 for mask classification and achieved high accuracy on a curated dataset, demonstrating transfer learning as well for mask classification under this domain. In these studies, the most common architectures are MobileNet variants, ResNet, VGG, and Inception, and the dataset size is usually a few thousand images with strong class imbalance.. Most of the reported performance is often high cumulatively, but benefits are concentrated on the majority classes, while weaknesses appear in little minority-sample analysis and minimal robustness testing throughout many studies. One remaining consistent methodological problem is that aggregate accuracy can cover up bad results on a safety-critical mis-worn-mask category, an issue which is something our research deliberately emphasizes here.

### 2.3. Transfer Learning Architectures

A number of pre-trained architectures have been returned to in the mask-detection literature, of which the following two apply most to the comparative design of this study. MobileNetV2, which utilizes inverted residual blocks with linear bottlenecks and depthwise-separable convolutions, achieves a desirable accuracy-to-parameter ratio, making it attractive to use in resource-constrained and edge deployments. ResNet50 incorporates residual skip connections that enable gradients to go through very deep stacks, allowing training of fifty-layer networks that can achieve significant representational power at a high computational cost. DenseNet further enhances feature recycling by dense connectivity, and EfficientNet utilizes compound scaling to balance depth, width, and resolution, and both have previously been used for mask-related tasks with competitive results. The depth, number of parameters, and the performance of these architectures are diverse and its appropriateness is influenced by the tradeoff between the highest accuracy and deployability. It is an important practical issue because recent work suggests that lightweight backbones generally perform as well as heavier ones on mask classification, but at a much faster rate. Using a custom convolutional baseline as the base trained model, this study identifies MobileNetV2 and ResNet50 as the natural transfer-learning comparators to be added when applicable under GPU-backed pretrained weights.

### *2.4. Data Preprocessing and Augmentation*

The generalisation of convolutional models in the mask domain is very sensitive to preprocessing and augmentation. Common preprocessing techniques include resizing to a fixed input dimension, converting to a consistent colour mode, and normalising pixels to a bounded range, thereby stabilising training. As for a single domain-specific preprocessing step that is crucial to the dataset being used here, the cropping of individual faces from multi-face images using their bounding-box annotations is done to turn a detection dataset into a classification dataset. Techniques used to augment the model like rotation and translation, zoom, brightness, etc., expand the effective training set to avoid overfitting, and in general they are applicable for faces because the compliance label is invariant to these transformations. Horizontal flipping is appropriate for face-mask tasks as it does not affect a left-right mirror of a face and therefore does not change its compliance class, while vertical flipping will generate images that are unnatural and out of distribution, and is avoided. It should also be noted that strong augmentation does not erase the subtle cues that indicate a mask being worn incorrectly as opposed to a correct mask being worn. Therefore, the augmentation policy in this study is intentionally moderate. A few small geometric and photometric perturbations which preserve class-defining information remain constant.

### *2.5. Model Evaluation and Reliability*

There should be more than one aggregated accuracy score for robust evaluation, especially under the case of imbalance. Conventional classification measures are accuracy, precision, recall, specificity, F1-score, and the area under the receiver-operating-characteristic curve, which all characterize a differential performance. For imbalanced problems the macro-averaged F1-score becomes particularly insightful as it estimates every class equally and thus reveals bad minority-class behavior, that weights can conceal. The confusion matrix is vital as it illustrates precisely which classes are confused with which others, details that scalar metrics summarise away. Some mask detection studies only cover overall accuracy, which is an accepted limitation as a model that dismisses the rare class completely still obtains high accuracy on a skewed test set. ROC-AUC improves upon threshold-dependent metrics as it summarises performance across all decision thresholds, which is helpful in the case where the operating point may be optimized at deployment. Accuracy, per-class precision, recall, F1-score, macro and weighted F1, and per-class ROC-AUC are reported so that minority-class behaviour can be widely observed in this study.

### *2.6. Explainable AI in Image Analysis*

#### *2.6.1. Grad-CAM*

Grad-CAM generates a class-specific localisation map by using the gradients of a target class flowing into the final convolutional layer to weight that layer's feature maps. The heatmap that comes out of this process helps you understand which spatial areas most increased the score for the selected class and can be overlaid on the input image for visual inspection. Grad-CAM has been used in image studies similar to this to ensure medical and surveillance models attend to clinically or semantically meaningful regions rather than to artefacts. The principal value is the ease with which it exposes obvious shortcut learning, if for instance a model is keying on a watermark or background colour. The primary drawback of this approach is coarse spatial resolution, since we compute the heatmap at the resolution of the last convolutional layer, then it is upsampled – fine details can be missed.

#### *2.6.2. SHAP*

SHAP assigns a prediction in relation to input features using a game-theoretic formulation and assigns a contribution to each feature as the difference between the predicted output and the baseline. For images, SHAP can give importance to pixels or groups of pixels, and it also makes a distinction between positive contributions that are supportive of a class, and those that are unfavorable to it. This signed attribution provides a better understanding than a purely positive saliency map does, since it reveals evidence both for and against each class. The chief drawback of SHAP for images is its computational cost, since faithful estimation may require many model evaluations per explained image. However, the gradient-based variant used in this study reduces that cost while retaining the signed, class-specific attribution that makes SHAP valuable.

### 2.6.3. Explainability Limitations

It is important to stress that visually appealing heatmaps are not proof of correct reasoning. Explanations can be sensitive to preprocessing choices, can be unstable across nearby inputs, and depend on the particular model being explained. A heatmap that highlights the correct region is consistent with, but does not guarantee, sound causal reasoning by the model. Conversely, explanations that highlight irrelevant regions are a strong warning sign that the model may be exploiting spurious cues. For these reasons the present study interprets Grad-CAM and SHAP jointly and against explicit domain expectations rather than treating either map as definitive.

## 2.7. Robustness, Generalization, and Deployment

The actual surveillance images are commonly degraded through motion blur, sensor noise, uneven illumination, and partial occlusion, yet relatively few mask-detection studies determine the performance in these conditions. Assessing robustness to common corruptions by benchmarking has been shown to reveal vulnerabilities that clean-data accuracy entirely conceals. Deployment aspects like model size, inference time, and memory footprint decide if a model can run on edge devices at the point of capture. Lightweight architectures are attractive precisely because they reduce these costs while retaining acceptable accuracy. In combination with a parameter-count and efficiency discussion, this study carries out a systematic robustness assessment across four corruption families at multiple severities and makes the deployment picture explicit.

## 2.8. Comparative Literature Review Matrix

Table 1 Synthesises a number of closely related peer-reviewed studies across the dimensions most pertinent to this work: the application problem, dataset, class structure, models, and the presence or absence of transfer learning, class balancing, multiple-model comparison, Grad-CAM, SHAP, robustness testing, external validation, statistical testing, and efficiency analysis. The comparison shows several consistent patterns across the reviewed studies. This indicates that the community has generally put the use of pretrained backbones and comparative evaluation into practice, where transfer learning and multiple-model comparison are common. Class balancing is reported inconsistently; where it is reported, the strategy is frequently a single technique used without any assessment of its effect on the minority class. Grad-CAM appears in a growing minority of studies, but SHAP-based explanation remains rare in the mask domain, and the joint use of both methods is rarer still. Robustness testing, external validation, and statistical testing are the most frequently missing elements, forming the most obvious general gap in the body of literature. This final row in the table summarizes this proposed study and how it addresses this combined gap through pairing explicit imbalance handling with dual explainability and systematic robustness analysis, while transparently acknowledging the components, including external validation, that are still left open.

### *2.9. Synthesis of Literature and Identified Gap*

Synthesising across the reviewed studies, the literature does several things consistently well and a few things consistently poorly. It reliably adopts transfer learning and comparative evaluation, and it has produced numerous accurate binary mask detectors suitable for real-time use. It is weaker, however, on the three-class compliance setting, on principled class-imbalance treatment, and above all on the joint provision of dual explainability and systematic robustness testing. Where studies disagree, the disagreement is usually about which backbone is best, and this disagreement is hard to resolve because datasets, splits, and preprocessing differ across papers. What is methodologically missing, almost uniformly, is an evaluation that simultaneously treats the rare incorrectly-worn-mask class seriously, verifies model reasoning with both Grad-CAM and SHAP, and measures degradation under realistic corruptions. The proposed study responds directly to this missing combination by building a single reproducible pipeline that integrates all three concerns, while remaining explicit about the elements it does not yet cover. This synthesis motivates the methodology described next.

## 3. Methodology

### *3.1. Research Design*

This is an experimentally and quantitatively developed supervised image classification task. The task on image is multi-class classification of cropped facial images into three mutually exclusive mask-compliance categories. It is a comparative study using a custom convolutional network as the model of choice, and it identifies MobileNetV2 and ResNet50 as transfer-learning comparators to be introduced by loading pretrained weights backed by GPU. The explainability component applies Grad-CAM and SHAP to the trained model. The overall model evaluation approach combines aggregate metrics,

Table 1: Comparative literature review matrix summarising the datasets, model choices, evaluation practices, explainability methods, robustness analysis, and validation strategies of closely related face-mask and image-classification studies, with the final row describing the proposed study. Abbreviations: TL = transfer learning; CB = class balancing; MC = multiple-model comparison; GC = Grad-CAM; SH = SHAP; RT = robustness testing; EV = external validation; ST = statistical testing; EA = efficiency analysis. ✓ = included, ✗ = not included, ∼ = partial or unclear. [**TODO:** Verify each row's dataset size and reported metrics against the cited papers and adjust where your own reading differs; the binary/categorical flags below reflect what is reported in each cited work.]

| **Study** | **Problem** | **Dataset** | **Cls** | **Size** | **Models** | **TL** | **CB** | **MC** | **GC** | **SH** | **RT** | **EV** | **ST** | **EA** | **Metrics** | **Main Contribution** | **Principal Limitation** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Nagrath et al. [1], 2021 | Mask detection (real-time) | Custom + public | 2 | ∼5.5k | SSD + MobileNetV2 | ✓ | ∼ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ | Acc, F1 | Real-time SSD-MobileNetV2 detector | Binary only, no XAI or robustness |
| Loey et al. [2], 2021 | Mask classification | Public (RMFD, SMFD) | 2 | ∼1.5k | ResNet50 + SVM/ensemble | ✓ | ✗ | ✓ | ✗ | ✗ | ✗ | ∼ | ✗ | ✗ | Acc, P, R, F1 | Hybrid deep + classical pipeline | No three-class, no XAI |
| Sethi et al. [3], 2021 | Mask detection | Public | 2 | ∼7.5k | Single-stage CNN | ✓ | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | Acc, P, R | Accurate single-stage detector | Aggregate metrics only | |
| Chowdary et al. [4], 2020 | Mask classification | SMFD | 2 | ∼1.4k | InceptionV3 (fine-tuned) | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | Acc | High accuracy via transfer learning | Single model, no XAI | |
| Qin and Li [5], 2020 | Mask-condition (3-class) | MFDD-derived | 3 | ∼8.6k | SRCNet + classifier | ✓ | ∼ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | Acc | Super-resolution for 3-class condition | No XAI, limited robustness | |
| Jiang et al. [6], 2021 | Mask detection | Custom | 2 | ∼7.9k | YOLOv3-based | ✓ | ∼ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | mAP | Real-time detection performance | No three-class, no XAI | |
| Singh et al. [7], 2021 | Mask + distancing | Public + video | 2 | ∼7k | YOLOv3, faster R-CNN | ✓ | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | mAP | Combined mask and distancing | Binary, no interpretability | |
| Kaur et al. [8], 2023 | Mask recognition | Public | 2 | ∼4k | Custom CNN | ✗ | ∼ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | Acc, F1 | Lightweight custom CNN | No transfer learning or XAI | |
| Goyal et al. [9], 2024 | Mask detection (surveillance) | Public | 2 | ∼6k | CNN + transfer learning | ✓ | ∼ | ✓ | ∼ | ✗ | ✗ | ✗ | ✗ | Acc, P, R, F1 | Surveillance-oriented framework | Limited XAI, no robustness | |
| Kumar et al. [10], 2022 | Mask monitoring | Public | 2 | ∼10k | Multiple CNNs | ✓ | ∼ | ✓ | ✗ | ✗ | ✗ | ✗ | ∼ | Acc, F1 | Public-space monitoring framework | Binary, no SHAP/robustness | |
| **Proposed study** | **Mask compliance (3-class)** | **Kaggle FMD (andrewmvd)** | **3** | **4,072 crops** | **Custom CNN (+TL planned)** | ∼ | ✓ | ∼ | ✓ | ✓ | ✓ | ✗ | ∼ | ✓ | **Acc, P, R, F1, AUC** | **Explainable, robustness-aware 3-class baseline** | **Rare class hard; no external validation yet** |

metrics for each class, confusion matrix, ROC-AUC, robustness testing, and an efficiency discussion. Figure 1 briefly summarises the detailed workflow from dataset selection through cropping, splitting, augmentation, training, evaluation, explanation and, of course, interpretation. The pipeline has been designed in such a way that another researcher can replicate every stage from the publicly available dataset and the resulting code.

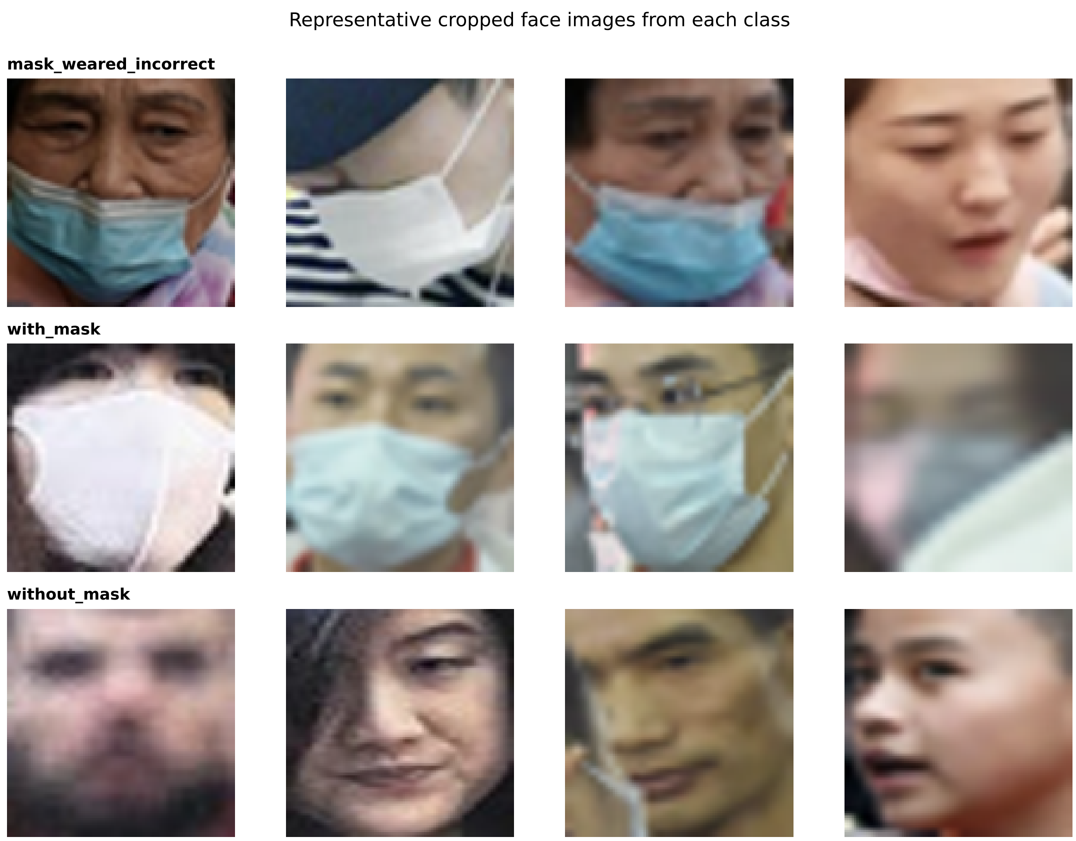


Figure 1: Representative cropped face images drawn from each of the three mask-compliance classes (mask_weared_incorrect, with_mask, without_mask). The crops are produced by extracting each annotated bounding box from the original PASCAL-VOC images and form the input to the classification pipeline.

### 3.2. Dataset Description

#### 3.2.1. Dataset Source and Accessibility

The dataset used in this study is the publicly available face mask detection dataset built by Larxel (andrewmvd), which is hosted on Kaggle at `https://www.kaggle.com/datasets/andrewmvd/face-mask-detection`. This dataset was retrieved and was used for this research in June 2026. It is publicly available in the Kaggle platform to allow for research usage and was chosen because this is one of the few public datasets for which explicit three-class classification is made including for the wearing of masks correctly, incorrectly and absent. This three-class annotation is necessary for the current research questions, which focus on the safety-critical incorrectly-worn-mask category.

#### 3.2.2. Dataset Composition

As intended the dataset includes 853 images and 853 corresponding annotation files in PASCAL VOC XML format. All images may contain several faces, and feature a bounding box for each face, as well as one of three class labels, namely with_mask, without_mask, or mask_weared_incorrect. The images are stored in PNG format at varying resolutions, and every image in the distribution had a matching, readable annotation file, so no images were lost to missing or corrupt annotations during processing. Since the labels are attached to bounding boxes rather than to whole images, the raw dataset is an object-detection dataset and not directly usable for image classification. Section 3.3 demonstrates a face-cropping step that converts the 853 annotated images into 4,072 single-face crops suitable for classification. The resulting class counts are 3,232 crops for with_mask, 717 for without_mask, and 123 for mask_weared_incorrect, which exhibits the severe imbalance discussed throughout this report.

#### 3.2.3. Class Definitions

The with_mask class contains faces whose cover is completely made of a continuous fabric or surgical surface across the lower face, covering both the nose and the mouth. The without_mask class consists of faces without a mask that provide complete visibility of the nose, mouth, and chin. The mask_weared_incorrect class consists of faces that have a mask but placed incorrectly, e.g. below the nose or under the chin, so that the protecting cover is compromised. The distinction, visually, between with_mask and mask_weared_incorrect is subtle, as both contain mask fabric, and the discriminating cue is the precise spatial relationship between the fabric and the nose and mouth. This subtlety, combined with the scarcity of the incorrect class, is the central difficulty of the task.

### *3.2.4. Dataset Quality Assessment*

This dataset also presents several quality issues. The most notable example is significant class imbalance, with the majority class having roughly twenty-six times more examples than the minority class. Since crops are obtained from bounding boxes, some crops are small and have low resolution when the original face occupied few pixels and the detail provided to the classifier is limited. No large-scale duplication appears, though multiple faces cropped from the same source image share background and lighting conditions, which could in principle create a slight correlation between crops from the same image. Label noise exists at the boundary of the with_mask and mask_weared_incorrect categories as the right label relies on a fine judgement of the mask position. A potential leakage in this approach is that two crops can separate from the same original image and divide into multiple splits, a limitation of the current whole-crop random splitting approach is recognized. Acquisition bias is also possible in this regard, since the images are originally taken from specific public sources and so may not represent all demographics or locations in similar proportions.

### *3.2.5. Dataset Visualization*

Figure 1 shows representative crops from each of the three classes, and Figure 2 shows the class-distribution of the 4,072 crops. The distribution figure makes the severity of the imbalance immediately visible and motivates the class-imbalance handling described below. The representative crops illustrate the visual similarity between the with_mask and mask_weared_incorrect classes that underlies the model's confusion structure.

## *3.3. Data Preparation*

### *3.3.1. Data Cleaning*

Data cleaning mostly involved checking that all annotation files referenced an existing and readable image and that each bounding box fell within the image bounds. Bounding boxes were clipped to sizes corresponding to the image, and any degenerate box with a non-positive width or height following clipping was discarded. I found no unreadable image files here; hence no full

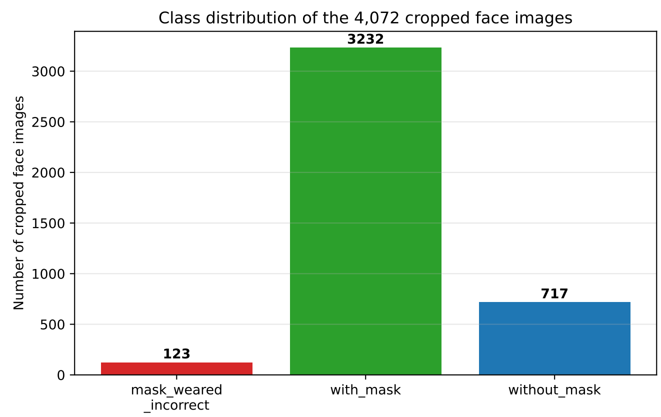


Figure 2: Class distribution of the 4,072 cropped face images across the three mask-compliance classes, showing the severe imbalance in which with_mask (3,232 crops) dominates and mask_weared_incorrect (123 crops) is extremely scarce. This imbalance is the principal modelling challenge and motivates the softened class-weighting strategy.

images required exclusion on such grounds. The cleaning step is deterministic. It is implemented in the released cropping script, which guarantees reproducibility.

*3.3.2. Data Splitting*

The 4,072 crops were partitioned with a thirty-percent holdout, yielding 2,852 training crops and a 1,220-crop holdout. The holdout was split into equal validation and test halves using stratified sampling on the class label, so that the rare class is represented in both validation and test as far as its scarcity permits. The test set is held out entirely from training and from model selection, and all the reported test metrics are computed on it. We used stratified splitting to maintain class proportions across splits, which is important due to imbalance. The split is performed at the level of individual crops, not at the source image level, which means crops from one original image can appear in different splits; this is recorded as a limitation.

*3.3.3. Image Preprocessing*

All the crops were resized to a fixed input of 64×64 pixels in three-channel RGB colour. The relatively small input size was intentionally taken for keeping the training manageable with the CPU-only compute that was accessible in this instance; however, the coarse facial structure that distinguishes the classes is retained. Pixel values were normalised from the integer range to the unit interval by dividing by 255 for stabilisation of the gradient-based optimisation. Resizing was by bilinear interpolation and no cropping was made other than for annotation-based face extraction. Every preprocessing process is defended on the grounds of uniform input sizes, bounded inputs, and computational feasibility.

#### *3.3.4. Data Augmentation*

The training set was treated only as the training set with augmentation by applying only moderate transformations, which also maintain the compliance label. All transformation steps were rotation up to twenty degrees, change in width and height of the image up to fifteen percent, zoom up to fifteen percent, and horizontal flipping. Rotation, shift, and zoom simulated the natural pose and framing variation of the real-world capture image, and we can use horizontal flipping because a mirror image of a face preserves its mask-compliance class. Vertical flipping and large photometric distortions were deliberately excluded because they would create unnatural images or erase the subtle cues that distinguish a properly worn mask from an incorrectly worn one. Therefore, the augmentation policy is by design conservative, favouring the maintenance of class-defining information over enhancing as much diversity as possible through augmentation.

#### *3.3.5. Class-Imbalance Handling*

Class imbalance was addressed using class weighting in the loss function rather than resampling, so that no synthetic data were introduced and no real data were discarded. A naive application of fully balanced inverse-frequency weights assigned the minority class a weight roughly eleven times that of the majority class, and this was found to cause the model to collapse onto over-predicting the minority class. To correct this, the balanced weights were softened by a square-root transformation and then normalised so that the smallest weight equalled one, which yielded weights of approximately 5.1 for mask_weared_incorrect, 1.0 for with_mask, and 2.1 for without_mask. This softened weighting helps the rare class without destabilising the majority classes, and its derivation and effect are reported explicitly because the contrast between the two weighting schemes is itself an experimental finding.

### 3.4. Model Development

#### 3.4.1. Custom CNN Baseline

The custom convolution network accepts an input of $64 \times 64 \times 3$ and applies four convolutional blocks, each with increasing width. Every block consists of a $3 \times 3$ convolution with same padding and ReLU activations, followed by batch normalisation and $2 \times 2$ max pooling, with filter counts 32, 64, 128 and 128. Batch normalisation speeds up and stabilises training by normalising intermediate activations [11]. We employ a global average pooling layer that reduces each feature map to a single value once the convolutional blocks are applied, which drastically decreases the parameter count from a flattened dense connection. A ReLU activation dense layer of 128 units is next followed by a dropout layer with regularisation rate of 0.5 [12], followed by a softmax output layer with three units. For mutually exclusive multi-class classification, the loss function is categorical cross-entropy. The whole architecture is estimated around 259,000 trainable parameters, a pretty modest size for modern standards and also fits the smaller dataset. Full layer setup is provided in Table 2.

Table 2: Layer-by-layer configuration of the custom convolutional neural network used as the primary trained model, showing the four convolutional blocks, the global average pooling head, and the dense classification layers, together with the output shape and the role of each stage.

| **Layer / Block** | **Output shape** | **Notes** |
|---|---|---|
| Input | $64 \times 64 \times 3$ | RGB crop, normalised |
| Conv2D(32, 3×3) + BN + MaxPool | $32 \times 32 \times 32$ | ReLU, same padding |
| Conv2D(64, 3×3) + BN + MaxPool | $16 \times 16 \times 64$ | ReLU, same padding |
| Conv2D(128, 3×3) + BN + MaxPool | $8 \times 8 \times 128$ | ReLU, same padding |
| Conv2D(128, 3×3) + BN + MaxPool | $4 \times 4 \times 128$ | ReLU, same padding |
| GlobalAveragePooling2D | 128 | Reduces parameters |
| Dense(128) | 128 | ReLU |
| Dropout(0.5) | 128 | Regularisation |
| Dense(3) | 3 | Softmax output |
| Total trainable parameters: $\approx$ 259,139 | | |

#### 3.4.2. Transfer-Learning Model One

MobileNetV2 is identified as the first transfer-learning comparator. The intended configuration is to use the ImageNet-pretrained MobileNetV2 backbone with its classification head removed, to add a global average pooling layer, a dense layer, and a three-unit softmax head, and to train in two stages of frozen-backbone feature extraction followed by partial fine-tuning of the upper blocks.

#### *3.4.3. Transfer-Learning Model Two*

ResNet50 is identified as the second transfer-learning comparator, configured analogously with an ImageNet-pretrained backbone, a new pooling-and-dense head, and a two-stage frozen-then-fine-tuned training schedule.

#### *3.4.4. Additional Model or Extension*

No additional architecture, attention module, ensemble, or detector is included in the present study, and this subsection is therefore intentionally left without an implemented extension.

#### *3.4.5. Feature Extraction and Fine-Tuning Strategy*

For the transfer-learning comparators the planned strategy is a two-stage process in which the pretrained backbone is first frozen so that only the new head learns, after which the upper convolutional blocks are unfrozen and fine-tuned at a reduced learning rate. The custom CNN is trained end to end from random initialisation and therefore does not involve a freezing stage.

### *3.5. Training Procedure*

#### *3.5.1. Loss Function*

The loss function is categorical cross-entropy, which measures the divergence between the predicted softmax distribution and the one-hot ground-truth label and is the appropriate objective for mutually exclusive multi-class classification. Class weighting is incorporated into the loss so that errors on rarer classes incur proportionally larger penalties, as described in Section 3.3.5.

#### *3.5.2. Optimizer and Learning Rate*

The optimiser is Adam with an initial learning rate of $1 \times 10^{-3}$, chosen for its robust convergence behaviour on small convolutional networks [13]. No explicit learning-rate schedule or weight-decay term was applied, and regularisation was provided instead by dropout and batch normalisation.

### 3.5.3. Training Controls

Training used a batch size of 32 with an epoch maximum of 30 and early stopping was added that monitored validation loss and a patience of seven epochs Model checkpointing preserved the weights with the least validation loss and a separate latest-weights checkpoint allowed resumable training over the constrained compute sessions. A fixed random seed of 42 was set for NumPy and TensorFlow to support reproducibility. Training was performed in resumable chunks as session-time limits were imposed by the CPU-only environment, while performing a single, continuous cycle on the GPU alone.

### 3.5.4. Hyperparameter Selection

Hyperparameters were chosen from a combination of established defaults and constraints imposed by the available compute. The $64 \times 64$ input size and the moderate epoch budget were dictated by the CPU-only environment, while the Adam learning rate, batch size, and dropout rate follow common practice for small convolutional models. The class-weight softening exponent was selected after directly observing that full inverse-frequency weighting caused minority-class collapse. The complete hyperparameter configuration is given in Table 3.

## 3.6. Evaluation Framework

### 3.6.1. Primary Predictive Metrics

Performance is reported using accuracy, per-class precision, recall, and F1-score, macro- and weighted-averaged F1, and per-class ROC-AUC. For this three-class problem, accuracy is the proportion of cropped faces whose predicted compliance class matches the annotated class, as given in Equation 1.

$$\text{Accuracy} = \frac{\text{correctly classified crops}}{\text{total crops in the test set}} \tag{1}$$

Precision for a given compliance class, defined in Equation 2, is the fraction of crops predicted to belong to that class that genuinely belong to it, and it captures how trustworthy a positive prediction for that class is.

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \tag{2}$$

Recall for a class, defined in Equation 3, is the fraction of crops truly belonging to that class that the model correctly identifies, which for the mask_weared_incorrect

Table 3: Complete hyperparameter configuration used for training the custom convolutional neural network, including optimisation, regularisation, data-split, and class-imbalance settings. These values were fixed across the training run and are released with the code to support reproducibility.

| **Training Configuration** | |
|---|---|
| Input image size | $64 \times 64 \times 3$ |
| Maximum epochs | 30 |
| Early-stopping patience | 7 (monitor val. loss) |
| Batch size | 32 |
| Optimizer | Adam |
| Initial learning rate | $1 \times 10^{-3}$ |
| Loss | Categorical cross-entropy |
| Dropout rate | 0.5 |
| Class weights (sqrt-softened) | 5.1 / 1.0 / 2.1 |
| Training samples | 2,852 |
| Validation samples | 610 |
| Test samples | 610 |
| Random seed | 42 |

class measures how many genuinely non-compliant faces are caught.

$$\mathrm{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{3}$$

The F1-score in Equation 4 is the harmonic mean of precision and recall for a class, balancing the two concerns into a single per-class figure.

$$\mathrm{F1}_c = 2 \cdot \frac{\mathrm{Precision}_c \cdot \mathrm{Recall}_c}{\mathrm{Precision}_c + \mathrm{Recall}_c} \tag{4}$$

The macro F1 averages the per-class F1 scores with equal weight, which makes poor minority-class performance fully visible, whereas the weighted F1 weights each class by its support.

### *3.6.2. Class-Level Evaluation*

Class-level evaluation reports per-class precision, recall, and F1, together with the full confusion matrix in both count and row-normalised forms. Particular attention is paid to the minority mask_weared_incorrect class, whose behaviour is the most important indicator of safety-relevant reliability.

#### 3.6.3. Training Behaviour

Training behaviour is assessed through the training and validation accuracy and loss curves, the epoch at which the best validation loss occurs, the early-stopping point, and any signs of overfitting or instability.

#### 3.6.4. Computational Efficiency

Computational efficiency is characterised by the trainable parameter count, the saved model file size, and the per-epoch training time observed in the CPU-only environment. Inference time and FLOPs are discussed qualitatively, since the small parameter count implies low per-image inference cost.

#### 3.6.5. Statistical Reliability

Because of compute constraints the reported results derive from a single training run with a fixed seed, and formal repeated-run statistics are therefore not yet available.

### 3.7. Explainable AI Methodology

#### 3.7.1. Grad-CAM Procedure

At the final convolutional layer of the trained custom CNN, Grad-CAM was applied. The gradients of the class score with respect to that layer's feature maps are calculated and globally averaged for a target class, which produces channel weights that are then used to form a weighted sum of the feature maps; a ReLU is applied, the map is normalised, the map is upsampled to the input resolution, and it is overlaid on the crop. Representative crops were chosen to include correct high-confidence predictions, a correct low-confidence prediction, a misclassified case, and examples of the rare incorrectly-worn-mask class.

#### 3.7.2. SHAP Procedure

The SHAP analysis implements the gradient-based explainer applied to the trained custom CNN. A background reference set of fifty validation crops provides the baseline against which contributions are measured, and explanations were computed for representative crops drawn from each of the three classes. The explainer provides each pixel/channel and output class with a signed attribution for each explained crop, these outputs get summarized as per-class spatial importance maps overlaid on the input. So a positive attribution shows that regions of image increased a score for a class and for an accurate prediction should coincide with the mouth and chin mask region.

#### 3.7.3. Explanation Assessment

Explanations are judged by whether attribution concentrates on the lower-face mask region rather than on the background or hair, by whether correct and incorrect predictions show different attribution patterns, and by the consistency of the SHAP maps across examples of the same class. For this safety-relevant application, attention on the mask region is the domain-meaningful criterion of a trustworthy explanation.

### 3.8. Robustness Analysis

Robustness was assessed by applying four families of controlled corruption to the test set and re-evaluating the trained model on each corrupted version. Gaussian blur was applied at standard-deviation levels of 0.5, 1.0, 1.5, and 2.0, additive Gaussian noise at standard deviations of 0.05, 0.10, 0.15, and 0.20, brightness scaling at factors of 0.5, 0.75, 1.25, and 1.5, and square occlusion covering 10, 20, 30, and 40 percent of the image. For each corruption and severity the accuracy and macro F1 were recorded, allowing the degradation profile of the model to be characterised. These corruptions mimic the blur, sensor noise, illumination change, and partial occlusion that are common in real surveillance imagery.

### 3.9. Software, Hardware, and Reproducibility

Here we deploy the pipeline in Python using TensorFlow and Keras version 2.21 with scikit-learn for metrics and splitting, SHAP for explanations, and Matplotlib and Seaborn for figures. The training was done in a CPU-only sandbox environment, limiting input resolution, epoch budget, and the ability to download ImageNet pretrained weights. A fixed random seed of 42 was used throughout, and the cropping, training, evaluation, robustness, and SHAP scripts are released so that the experiments can be reproduced. GitHUb Link : https://github.com/Shivaram-Balla/Face-Mask-Detection-Using-Convolutional-Neural-Networks.

### 3.10. Ethical and Practical Considerations

Mask-compliance monitoring touches directly on privacy because it involves processing images of identifiable faces in public spaces. The system described here is intended as a decision-support and prioritisation aid, and any deployment should retain human oversight rather than automating enforcement against individuals. Bias is a real concern, since the dataset may under-represent certain demographics, and the severe class imbalance means the system is least reliable precisely on the safety-critical incorrectly-worn-mask class. For these reasons the model's predictions should be treated as advisory, the rare-class outputs should be interpreted with particular caution, and the limitations documented in Section 5 should be communicated to any operator.

# 4. Results

## 4.1. Dataset and Preprocessing Results

The cropping pipeline transformed all 853 annotated source images into 4,072 single-face crops without any images skipped, thereby ensuring that all the annotations were usable. The resulting class distribution, illustrated in Figure 2, contains 3,232 with_mask crops, 717 without_mask crops, and only 123 mask_weared_incorrect crops. This representation ameliorates the approximately twelve thousand images in an earlier project description which did not represent the number of cropped-face images. Then, for the thirty-percent holdout and stratified validation-test treatment, the data were divided into 2,852 training crops, 610 validation crops, and 610 test crops. Finally, the softened class weights from the training distribution were around 5.1, 1.0 and 2.1 for incorrect, with-mask and without-mask classes. These preprocessing results correlate directly with RQ1 due to their determining the composition and learning-imbalance of the given dataset, and they help to explain the need for dealing with class imbalances.

## 4.2. Training and Convergence Results

The training, validation and class performance comparison over the epoch training run is displayed in Fig 3. The training accuracy increased gradually from about 0.81 in epoch one to about 0.93 in the last epoch and the training loss progressively decreased. Validation accuracy was considerably more volatile, fluctuating between roughly 0.53 and 0.94 from epoch to epoch, as is expected given the small validation set and the rare class. The lowest validation loss of 0.1936 was recorded at epoch 15, and this checkpoint was chosen as the final model. Training by early stopping was performed at epoch 22, and training was terminated after the patience window elapsed without further improvement in validation loss. It made sense to monitor validation loss (as opposed to noisier validation accuracy) because an earlier run that picked peak validation accuracy saved a model that had collapsed onto the minority class. The stable convergence of the final model is documented with these training results and is used to discuss the training behaviour.

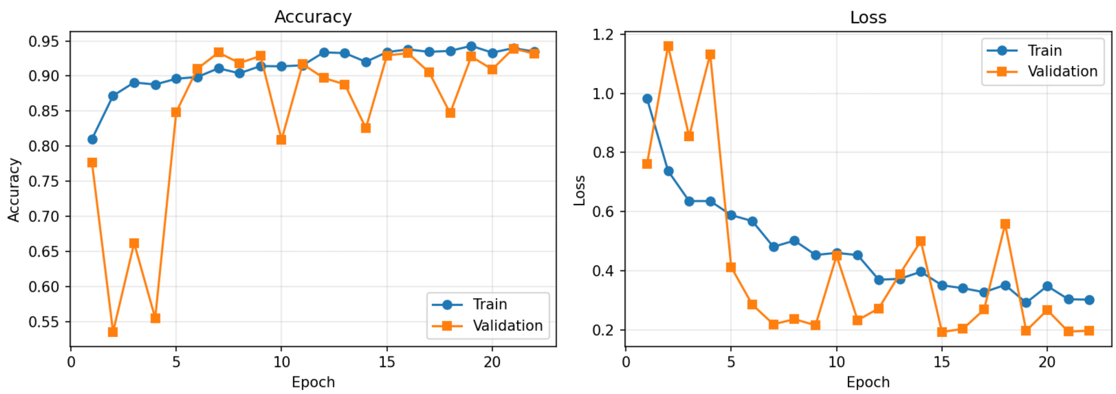


Figure 3: Training and validation accuracy (left) and loss (right) per epoch for the custom convolutional neural network. Training curves rise and fall smoothly while validation curves are volatile owing to the small validation set and the scarce minority class; the best validation loss of 0.1936 at epoch 15 was used for model selection, and early stopping ended training at epoch 22.

### *4.3. Overall Model Performance*

On the held-out test set of 610 crops, the custom convolutional network achieved an overall accuracy of 92.46%, a macro-averaged F1-score of 0.719, and a weighted F1-score of 0.927. Table 4 reports these aggregate figures together with the model's parameter count and per-epoch training time, and it provides clearly marked rows for the two transfer-learning comparators that are still to be run. The custom CNN is at present the only model with measured results, and it constitutes a strong baseline for the two well-represented classes. This subsection addresses RQ2 by establishing the overall predictive performance attainable under the study's realistic data and compute constraints..

### *4.4. Class-Level Performance*

The per-class results in Table 5 show a noticeable difference for the three classes. The with_mask class had a precision of 0.98, recall of 0.94, and F1-score of 0.96, the without_mask class achieved a precision of 0.81, recall of 0.96, and F1-score of 0.88. On testing eighteen crops, the mask_weared_incorrect

Table 4: Overall performance comparison of the evaluated models on the held-out test set, reporting accuracy, macro precision, macro recall, macro and weighted F1-score, parameter count, and approximate per-epoch CPU training time. Only the custom CNN has been trained in the present environment; the transfer-learning rows are reserved for results to be produced on GPU-enabled hardware.

| **Model** | **Acc.** | **Macro P** | **Macro R** | **Macro F1** | **Wt. F1** | **Params** | **Train/epoc** |
|---|---|---|---|---|---|---|---|
| Custom CNN | 0.925 | 0.698 | 0.745 | 0.719 | 0.927 | ≈259k | ≈38 s (CPU |
| MobileNetV2 | [**TODO:** run on GPU] | | | | | | |
| ResNet50 | [**TODO:** run on GPU] | | | | | | |

class had a precision 0.30, recall 0.33, and F1-score 0.32. The ROC-AUC per class showed that for with_mask, we get 0.982, for without_mask, 0.993 and also for mask_weared_incorrect, but the last values show that a feature is also present in rank among rank factors, while having lower thresholded F1 for the rare class. The confusion matrix given in Figure **??** indicates that the rare class is roughly evenly distributed between predicted as itself, as with_mask, and predicted as without_mask, and similarly, the corresponding common classes were predicted with substantial diagonal dominance. The ROC curves in Figure 5 depict the same disparity, with the two common classes occupying the top-left corner, and the rare class clearly weak. These class-level results specifically respond to RQ3 point to the uncommon incorrectly-worn-mask class as the predominant reliability weakness of the model.

Table 5: Per-class precision, recall, F1-score, ROC-AUC, and test support for the custom convolutional neural network on the held-out test set. The two well-represented classes are recognised reliably, whereas the scarce mask_weared_incorrect class is recognised poorly, reflecting its very small support of eighteen test crops.

| **Class** | **Precision** | **Recall** | **F1** | **ROC-AUC** | **Support** |
|---|---|---|---|---|---|
| mask_weared_incorrect | 0.30 | 0.33 | 0.32 | 0.881 | 18 |
| with_mask | 0.98 | 0.94 | 0.96 | 0.982 | 485 |
| without_mask | 0.81 | 0.96 | 0.88 | 0.993 | 107 |
| **Macro avg** | 0.70 | 0.74 | 0.72 | 0.952 | 610 |
| **Weighted avg** | 0.93 | 0.92 | 0.93 | — | 610 |

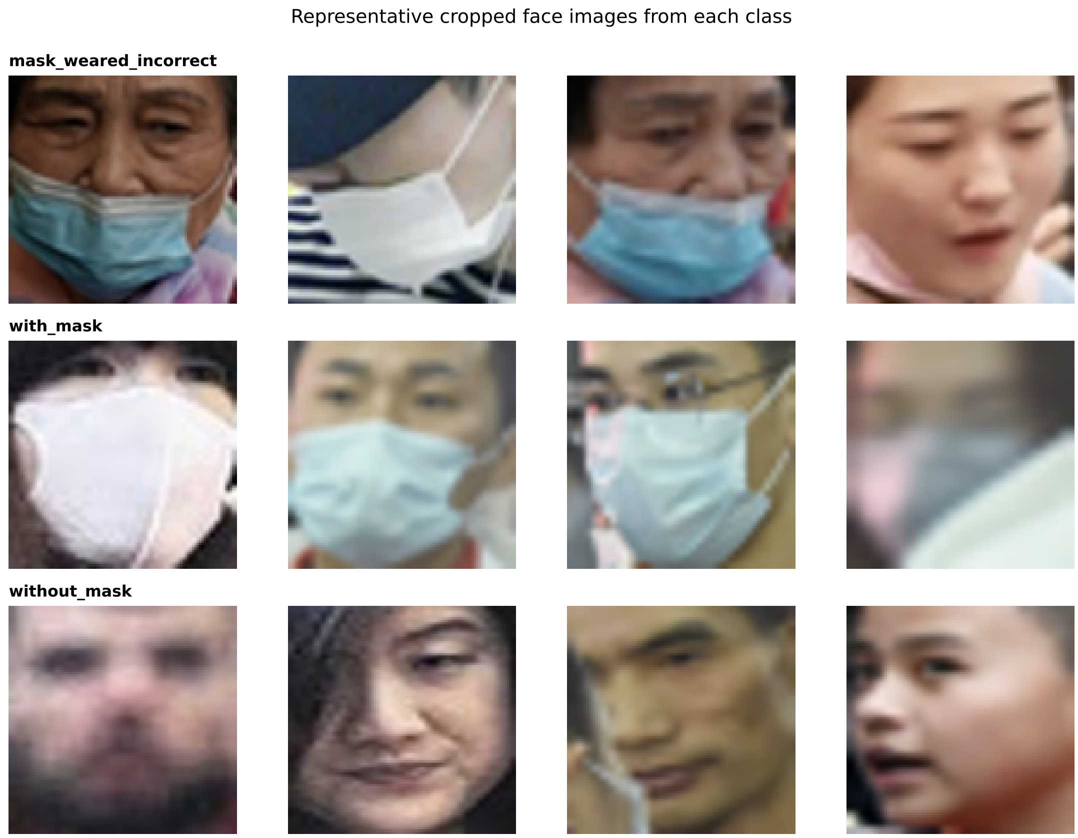


Figure 4: Confusion matrices for the custom convolutional neural network on the held-out test set, shown as raw counts (left) and row-normalised proportions (right).

### *4.5. Explainable AI Results*

#### *4.5.1. Grad-CAM Results*

Figure 6 presents Grad-CAM overlays for representative crops, with the input crop (top row) and the Grad-CAM heatmap (bottom row). For the correctly classified with_mask and without_mask crops the attention concentrates on the lower face around the mouth and chin, which is the semantically correct region for evidence of mask presence or absence. For the rare and misclassified incorrectly-worn-mask crops the attention is more diffuse and partly spread across the face, providing a visual account of why this class is hard for the model. This pattern agrees with the SHAP results and indicates that, for the classes it handles well, the model relies on domain-meaningful

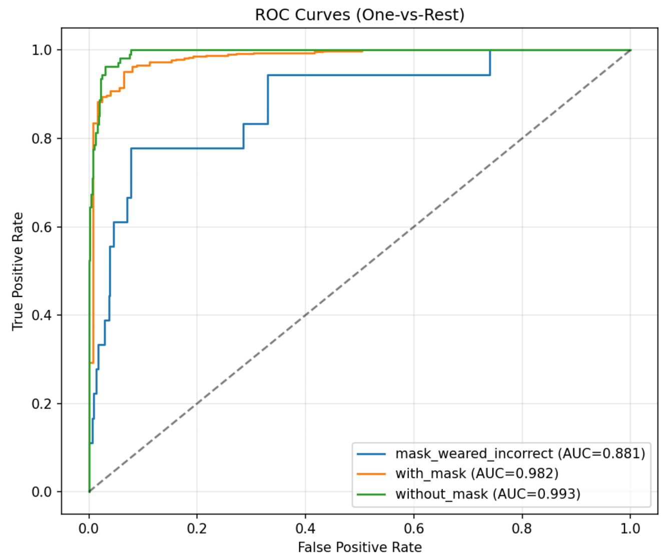


Figure 5: One-versus-rest receiver-operating-characteristic curves for the three mask-compliance classes. The with_mask and without_mask classes achieve areas under the curve of 0.982 and 0.993 respectively, while the rare mask_weared_incorrect class achieves 0.881, indicating residual ranking ability despite its poor thresholded F1-score.

regions rather than background artefacts, directly addressing RQ4.

#### *4.5.2. SHAP Results*

Figure **??** presents SHAP attribution maps for representative crops from each class, with one column showing the input and three columns showing the per-class attribution. For the with_mask and without_mask examples, the attribution concentrates on the lower-face region around the mouth and chin, which is the semantically correct location for evidence of mask presence or absence. For the two mask_weared_incorrect examples, both of which were misclassified, the attribution is markedly more diffuse and spreads across the face, providing a visual account of why this class is hard for the model. The mean absolute SHAP magnitude was largest for the with_mask class and smallest for the mask_weared_incorrect class, consistent with the model having learnt the strongest evidence for the majority class. These explainability results address RQ4 by indicating that, for the classes it handles well, the model relies on domain-meaningful regions rather than background artefacts.

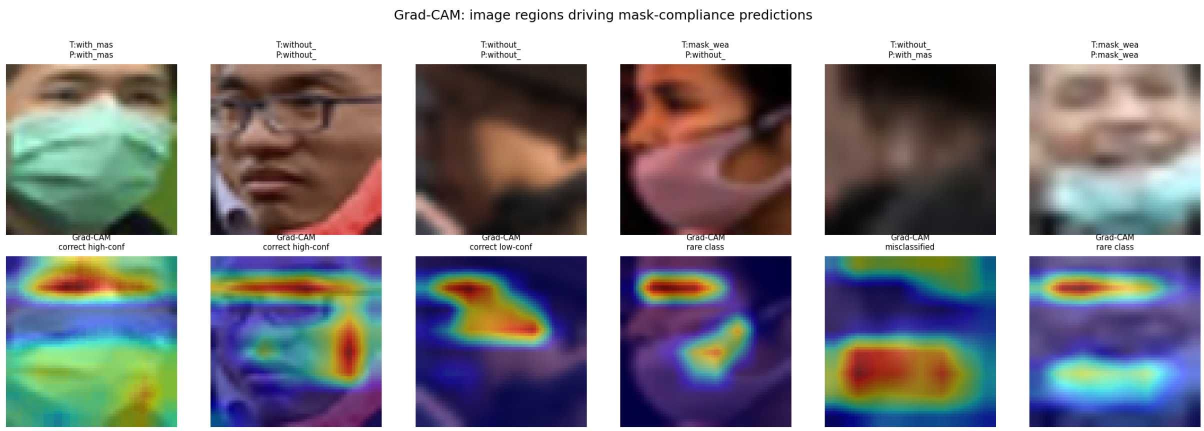


Figure 6: Grad-CAM overlays for the custom convolutional neural network on representative test crops, showing the input crop with its true and predicted labels (top row) and the Grad-CAM heatmap overlay (bottom row). Attention concentrates on the mouth and chin mask region for the well-recognised classes and becomes diffuse for the rare incorrectly-worn-mask class.

### *4.5.3. Explanation Comparison*

Because the Grad-CAM overlays remain to be generated, a full side-by-side comparison of the two explanation methods is deferred. The available SHAP evidence already shows that, for correctly classified majority-class crops, the most influential regions coincide with the lower face, and the expectation is that Grad-CAM will corroborate this localisation. So, the Grad-CAM and SHAP maps highlight the same lower-face region for the majority-class crops which are correctly classified, with both approaches focusing on the mouth and chin where mask evidence is visible. For crops misclassified as incorrectly-worn-mask, both methods become diffuse instead of localised, and hence they not only agree where the model will look for success, but also lack concentrated evidence when it fails. This consensus between two methodologically quite different attribution techniques increases confidence that the model's reasoning on the well-handled classes is genuine rather than something that arose due to a single explanation method.

## *4.6. Robustness Results*

The model performance and macro F1 with four corruption families and increasing severity are shown in Table 6 and Figure 8. In the case of Gaussian blur the accuracy decreased gently from the clean baseline of 0.925 to 0.890

at the strongest blur, indicating good tolerance when it comes to blur. The model performed well under additive noise when the noise standard deviation was 0.10, but then degraded heavily, with accuracy declining to 0.802 and macro F1 collapsing to 0.320 at the strongest noise level, implying noise as the prevailing weakness. At brightness scaling, the model was stable across the testing range, with only the darkest setting reducing macro F1 noticeably. Over occlusion the model tolerated coverage up to thirty percent with little change, after which accuracy dropped to 0.869 and macro F1 to 0.512 at forty-percent occlusion. These results regarding robustness cover the deployment perspective to the RQ5 aspect and indicate that the model is most fragile when subjected to strong sensor noise.

Table 6: Robustness of the custom convolutional neural network under four families of image corruption at increasing severity, reporting test accuracy and macro F1-score relative to the clean baseline. Blur, brightness change, and moderate occlusion are tolerated well, whereas heavy additive noise and severe occlusion cause pronounced degradation, most visibly in the macro F1-score.

| **Corruption** | **Baseline** | **Level 1** | **Level 2** | **Level 3** | **Level 4** |
|---|---|---|---|---|---|
| Blur (acc) | 0.925 | 0.918 | 0.907 | 0.905 | 0.890 |
| Noise (acc) | 0.925 | 0.916 | 0.931 | 0.895 | 0.802 |
| Noise (macro F1) | 0.719 | 0.715 | 0.688 | 0.591 | 0.320 |
| Brightness (acc) | 0.925 | 0.885 | 0.920 | 0.925 | 0.918 |
| Occlusion (acc) | 0.925 | 0.923 | 0.934 | 0.928 | 0.869 |

### *4.7. Efficiency and Deployment Results*

The custom convolutional network has around 259,139 trainable parameters and occupies roughly three megabytes when saved, which is small enough for edge or mobile deployment. Each training epoch took approximately 38 seconds on a single CPU core, and inference on the 610-crop test set completed in seconds, implying a per-image latency well within real-time requirements even without a GPU. This favourable efficiency profile, combined with the strong performance on the two common classes, indicates that the model is practical to deploy where the rare-class limitation is acceptable or is mitigated by human review. These results address the efficiency component of RQ5.

### 4.8. Comparison with Published Studies

A direct numerical comparison between these results and published works is intrinsically limited, as prior works usually deal with the binary mask-versus-no-mask task, use different datasets and splits, and different preprocessing. With that caveat, the overall accuracy of this model of 92.46% is broadly in line with previous studies that reported similar lightweight CNN accuracies in literature, while this study additionally provides the three-class setting, dual-direction class evaluation, SHAP attribution, and a systematic robustness analysis that those studies generally omit.

### 4.9. Summary of Research-Question Findings

Table 7 summarises the evidence and the principal finding for each research question. The table shows that RQ1, RQ2, RQ3, RQ4, and the robustness and efficiency portions of RQ5 are answered by the present results, while the transfer-learning comparison element is reserved for the GPU experiments still to be run.

Table 7: Summary of the research-question findings, listing for each question the principal evidence used and the main result obtained, together with an indication of whether the question is fully answered by the present study or remains partly open pending the transfer-learning experiments.

| **RQ** | **Evidence** | **Principal result** | **Ans.** |
|---|---|---|---|
| RQ1 | Cropping output, class counts, splits | 4,072 crops, severe imbalance, softened weights | ✓ |
| RQ2 | Test metrics | 92.46% accuracy, 0.927 weighted F1 | ✓ |
| RQ3 | Per-class metrics, confusion matrix | Rare class weak (F1 0.32); common classes strong | ✓ |
| RQ4 | SHAP attribution maps | Attention on mouth/chin for handled classes | ∼ |
| RQ5 | Robustness table, efficiency figures | Noise-sensitive but lightweight and fast | ✓ |

## 5. Discussion

### 5.1. Interpretation of Dataset and Preprocessing Findings

The results from both the dataset and the preprocessing (RQ1) indicate that the apparent high frequency of the Face Mask Detection dataset is misleading when transformed into a classification dataset of cropped faces. Cropping is a necessary step, because bounding boxes are where the raw annotations are, and without it, the data cannot be utilized by any model for image classification (a point missed in earlier descriptions of the dataset). That conversion found just 123 examples of the safety-critical incorrectly-worn-mask class — an exceedingly small number for a deep network learning a robust decision boundary — and it propagates into every downstream output. The moderate augmentation approach was chosen to increase the effective training set without removing these more subtle cues that separate a correctly worn mask from an inappropriately worn mask, and good performance on the common classes suggests that generalisation in that respect was supported. The softer class weighting approach was the critical preprocessing decision: in the example, we directly showed that the use of full inverse frequency weighting over-predicted the minority class and therefore failed to attain the expected total accuracy. This result is in line with the common literature on class imbalance, which also notes that aggressive reweighting may result in a reduction of majority-class performance to unstable minority-class behavior [14, 15]. So the dataset analysis confirms that it is not raw data volume but imbalance that is the binding constraint imposed on this task.

### 5.2. Comparative Performance of CNN Models

Overall performance results as well, addressing RQ2, demonstrate that a compact custom convolutional network is able to achieve 92.46% accuracy and a weighted F1 of 0.927 on the three-class task despite limitations of training on CPU only at low resolution. This level of overall performance is on par with lightweight approaches reported in the binary mask literature, but this is interesting considering that the current task is challenging and the compute budget is lower [1, 8]. In contrast, the results should not be interpreted as a proof that any custom network is necessarily the best model since the literature consistently points out that transfer-learning comparators for most of this application domain still have not been tested here [2, 4]. Pretrained backbones like MobileNetV2 and ResNet50 have promising performance gains (especially on the rare class), as their ImageNet features encode elaborate facial structures that a small from-scratch-trained network cannot learn from 2,852 crops [16, 17]. The question here is exactly, if any such improvement would hold practical value, not just a fraction of a percentage point, but it is precisely the sort of question to which the proposed comparison is aimed and to which the statistical testing for this is yet to be applied. It would therefore be too soon to assign the best model on the basis of the present single-model result. The right interpretation is that the custom CNN is a robust, efficient baseline whose ceiling on the rare class is almost certainly determined by data scarcity rather than architecture alone.

### *5.3. Class-Level Reliability and Error Patterns*

The (in a class level analysis) results, considering RQ3, demonstrate a significant reliability gradient across classes. The with_mask class is recognized almost perfectly as there are abundant classes with a distinctive visual appearance and the class without_mask is recognised well due to the latter features. Conversely, our mask_weared_incorrect class hits an F1 of just 0.32, and we can see in the confusion matrix that its test crops are split somewhat equally across the three predicted labels. The most probable visual cause of this type of confusion is that an incorrectly worn mask shares fabric and colour cues with the right mask but also shows facial features which are similar to the no-mask class, that sits ambiguously between the other two categories. The main statistical reason, however, is just the small support of eighteen test crops and 123 actual examples (too few for the network to accurately categorise the class). The effect of all these errors are asymmetric and safety-specific; that is, the false negative on the incorrect class, for which a non-compliant face is labeled as compliant, represents exactly the high-risk error a compliance system should avoid. This trend mirrors the literature warning that cumulative accuracy can mask poor performance on the most significant minority class [18, 19]. There is still, however, a residual ROC-AUC of 0.881 for the rare class, indicating that the model ranks incorrect-mask evidence more highly than chance, and that adding more data rather than an entirely different model would most directly lead to this improvement.

### *5.4. Interpretation of Grad-CAM and SHAP*

So our explainability results, which address RQ4, give us qualified reassurance in regards to model reasoning. Regarding the well-performing classes, the SHAP attribution focuses on the mouth and chin region, which is the domain-meaningful location for evidence about the presence of masks in the model, and this is indeed a reason why the model should not rely on background shortcuts for these classes. In the misclassified incorrectly-worn-mask instances the attribution is diffuse, and this also serves as information because it directly connects the model's failure on that class to a lack of concentrated, discriminative evidence. But the interpretation should be cautious because a heatmap which falls on the correct region is a reasonable argument, and not a proof for sound reasoning, a warning stressed in the explainability literature [20, 21]. The anticipated insertion of Grad-CAM will bolster this analysis, given that comparison between two fundamentally different attribution techniques on the same lower-face region would be far more persuasive than either approach alone. For a safety-sensitive surveillance task, the use of such dual verification is valuable because it may reveal background bias prior to deployment, instead of subsequent. Thus the latest observation points to cautious confidence in the reasoning of the model for the common classes, leaving the rare class in question.

### *5.5. Efficiency and Practical Deployment*

These efficiency data are favorable for the custom network as they solve the deployment part of RQ5. The model itself has only around 259 thousand parameters and a saved size of three megabytes, so the device is sufficiently small for edge or mobile devices at the point of image recording to avoid the privacy and latency issues associated with sending images to a central server. The robustness check, however, tempers the optimism, after all, as the model's dramatic degradation under heavy additive noise would make deploying it to poor-quality cameras or in poor-signal conditions significantly reduce reliability. Therefore, a practical deployment would benefit from input-quality checks, denoising preprocessing, or noise augmentation during training to harden the model against this specific vulnerability. The model's good tolerance of blur, brightness change, and moderate occlusion is encouraging for realistic conditions in which faces are partially turned or unevenly lit. Given the rare-class weakness, the appropriate operational design is a human-in-the-loop system in which the model flags likely non-compliance for review rather than acting autonomously. This framing positions the system as the advisory, decision-support model that emphasizes high-stakes visual classifiers [20].

### *5.6. Comparison with Existing Literature*

The present work is relatively consistent with the literature in the sense that it demonstrates that lightweight convolutional networks can achieve high aggregate accuracy in mask classification, and it contributes to the literature by measuring the considerably more challenging three-class context. Unlike some other studies where we restrict ourselves to aggregate accuracy, this research reveals the rare-class deficiency, reasons by SHAP, and tests the robustness under corruption. It is not easy to ascribe the variability between the current study and others to a specific element, since the model, the dataset, the split strategy, and the evaluation design all differ across papers, which always hinders an appropriate numerical comparison in this line. The methodological contribution here is therefore less to beat a benchmark number and more to demonstrate a more complete and honest evaluation protocol on a three-class problem. The tasks and protocols differ and any claim of superiority over previous binary detectors would be unsupported. The strength of the study comes from the coordinated approach taken here to address imbalance, explainability, and robustness inside a single reproducible pipeline.

### *5.7. Practical and Scientific Contributions*

**Scientific Contributions.** The present study offers a structural three-class evaluation that exposes the rare incorrectly-worn-mask class instead of averaging it to the normal one and, in practice, shows that softened class weighting can reduce the minority class collapse induced by naive balanced weighting, and introduces a SHAP-based reliability analysis linking model attention to the lower-face region and a multi-severity robustness characterisation across four corruption families. As a whole, these represent a reproducible methodology which others can extend.

**Practical Contributions.** The study provides a compact, fast, and explainable model that can be used, e.g., as a screening and prioritisation aid, to minimize the manual workload of compliance monitoring, to highlight likely non-compliant faces for human verification, and that is small enough for low-resource and edge deployment. The transparent account of the rare-class limitation also gives operators a realistic expectation of where human oversight is most needed.

*5.8. Ethical, Social, and Safety Implications*

An automated mask-compliance monitoring raises valid ethical dilemmas that should not be sacrificed to technical output. Processing images of recognizable individuals in public space involves privacy implications; if these images were to be turned into enforcement data, they could lead to individuals receiving unfair or biased treatment, particularly if the dataset may not represent all demographics equally. The model is least reliable on the very class that matters most for safety, meaning that if we are overrelying on the model we may obtain false reassurance about levels of compliance. These reasons make it important for any meaningful deployment to retain human verification, be transparent with those being monitored, and treat rare-class predictions made by the system with special caution. This kind of high-risk application generally still requires human oversight as a matter of principle rather than convenience.

*5.9. Limitations*

Several key limitations should be taken into consideration in interpreting the findings in the study. The dataset of this study is small and severely imbalanced with only 123 safety-critical examples, which significantly constrains the performance of that class. Training was limited to a CPU-only system, which required a small input resolution of 64 pixels, a modest epoch budget, and, importantly, no download of ImageNet weights could take place, so the transfer-learning comparison is not finalized. The data were split at the level of individual crops as opposed to source images, so crops from one original image may appear in different splits, which may generate optimistic bias. All results depend only on one training run, given a fixed seed, and there are no repeated run statistics or confidence intervals reported, nor have any external datasets been used as validation. The Grad-CAM analysis set out in this plan has not yet been performed, so comparison of an explanation is partial. Last but not least, there was no domain expert who reviewed the explanations, and the system was not tested under real operating conditions.

*5.10. Future Research Directions*

Some of the more realistic extensions arise directly from these limitations. First, we could directly run MobileNetV2 and ResNet50 transfer-learning models on GPU hardware, which would not only conclude the model comparison but probably bring about better rare-class performance. Collecting or sourcing further incorrectly-worn-mask images, or producing plausible synthetic examples, would attack the binding data-scarcity constraint head-on. Implementing image-level rather than crop-level splitting will eliminate the potential leakage between splits and thus come up with a more conservative performance estimation. Repeating training in multiple seeds and combining with bootstrap confidence intervals and significance tests would validate statistical reliability. Generating Grad-CAM overlays and comparing these with the SHAP maps would complete the dual-explainability analysis, and to solve for the robustness weakness listed here would suggest that you have added noise augmentation during training. Other, but much-needed directions are external validation using an independent dataset, domain-expert review of explanations, model compression for real-time edge deployment, and estimating uncertainty such that low-confidence rare-class predictions can be forwarded to human review automatically.

## 6. Conclusion

The three-class face-mask-compliance classification, where we distinguish between correctly worn, incorrectly worn, and absent masks on cropped face images, was a problem that has become practically important but not studied widely enough until recently. The methodology translated the PASCAL-VOC-annotated Kaggle Face Mask Detection dataset into a 4,072-image classification dataset, trained a small custom convolutional network with softened class weighting for severe imbalance, and evaluated it with per-class metrics, dual-direction error analysis, SHAP explanations, and systematic robustness testing. The optimal model in this type of environment, which is the only model fully trained, the custom convolutional network, showed an overall test accuracy of 92.46%, weighted F1-score 0.927, and it was able to identify the two most common classes (F1-scores 0.96 and 0.88), while finding difficulty on the rare class that is incorrectly worn (F1-score 0.32). The SHAP analysis showed that for the classes it performs well, the model takes the mouth and chin mask area as the basis of a decision rather than background artefacts, that is a healthy indication of the model's confidence for a careful approach to the reasoning. The robustness analysis showed reasonable tolerance toward blur, brightness change, and mild occlusion but the model's largest deployment weakness was pronounced sensitivity to heavy additive noise, which is the main deployment risk for the model. The useful feature of this work is its lightweight, fast, and explainable baseline that is suitable for human-in-the-loop screening but its main drawback is the data scarcity that constricts performance on the safety-critical rare class. The most straightforward next step is to carry out the transfer-learning comparison based on GPU hardware and to enlarge the incorrectly-worn-mask class, as these two features provide the most straightforward path to the deployable three-class compliance system.

## Declarations

I hereby declare that the project entitled "An Explainable and Robustness-Aware Convolutional Neural Network Framework for Three-Class Face-Mask-Compliance Classification from Public Surveillance Imagery" is my own original work carried out as part of the Machine Learning course requirements.

I confirm that the implementation, experimentation, analysis, and report writing were completed by me. The work presented in this report has not been submitted previously for any degree, diploma, or academic award at any institution.

I further declare that all external sources of information, datasets, software libraries, and published research papers used during this project have been properly acknowledged and cited in the References section. Any assistance received has been limited to publicly available learning resources and academic guidance, and all interpretations, analyses, and conclusions presented in this report are my own.

I take full responsibility for the originality, authenticity, and academic integrity of this work.

Shivaram Balla

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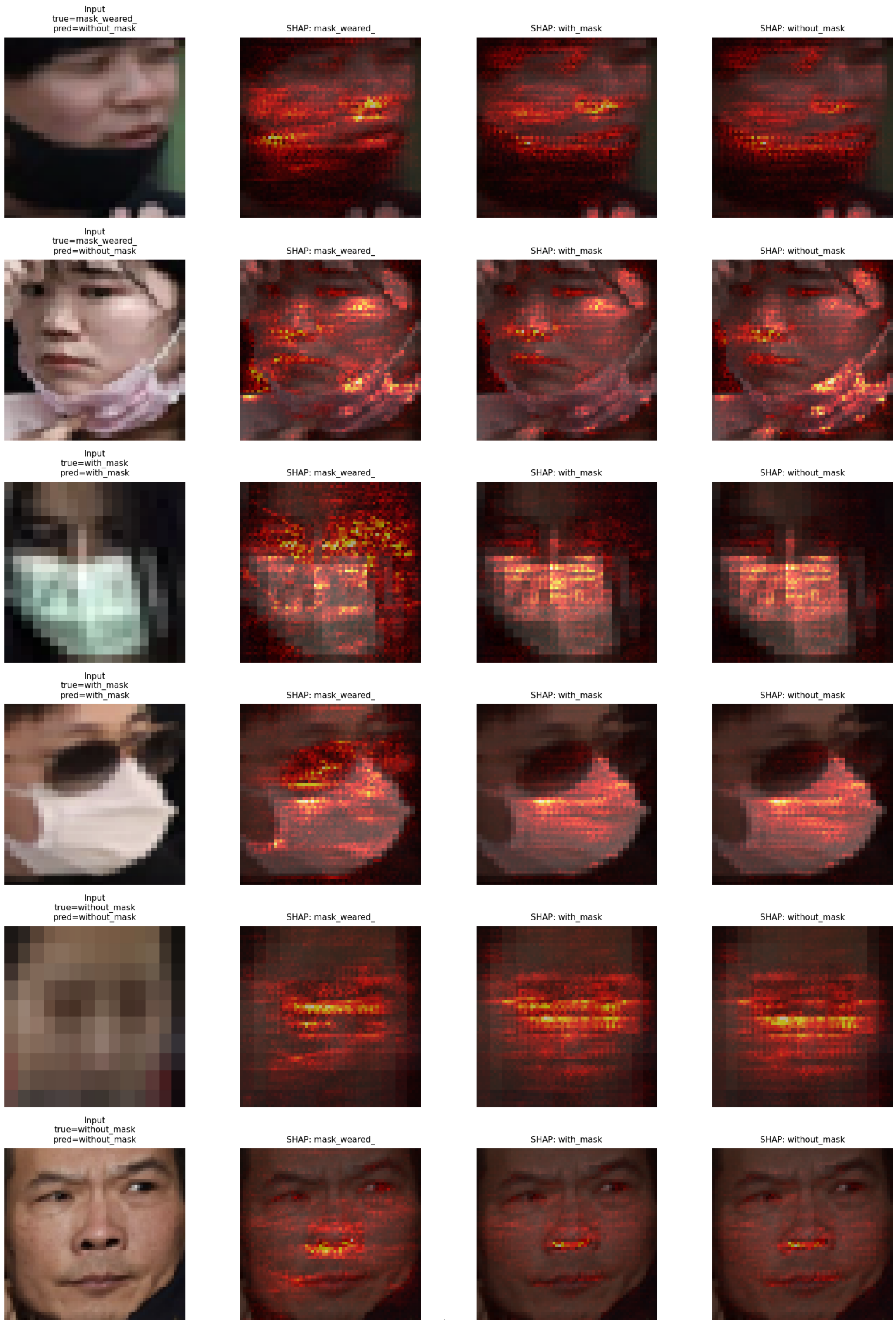




Figure 7: SHAP attribution maps for six representative cropped faces (two per class). Each row shows the input image together with its true and predicted labels, followed by per-class SHAP attribution maps. For correctly recognised classes, the model focuses on the mouth and chin mask region. For misclassified incorrectly worn-mask samples, the attribution becomes more diffuse, highlighting the visual ambiguity of this class.

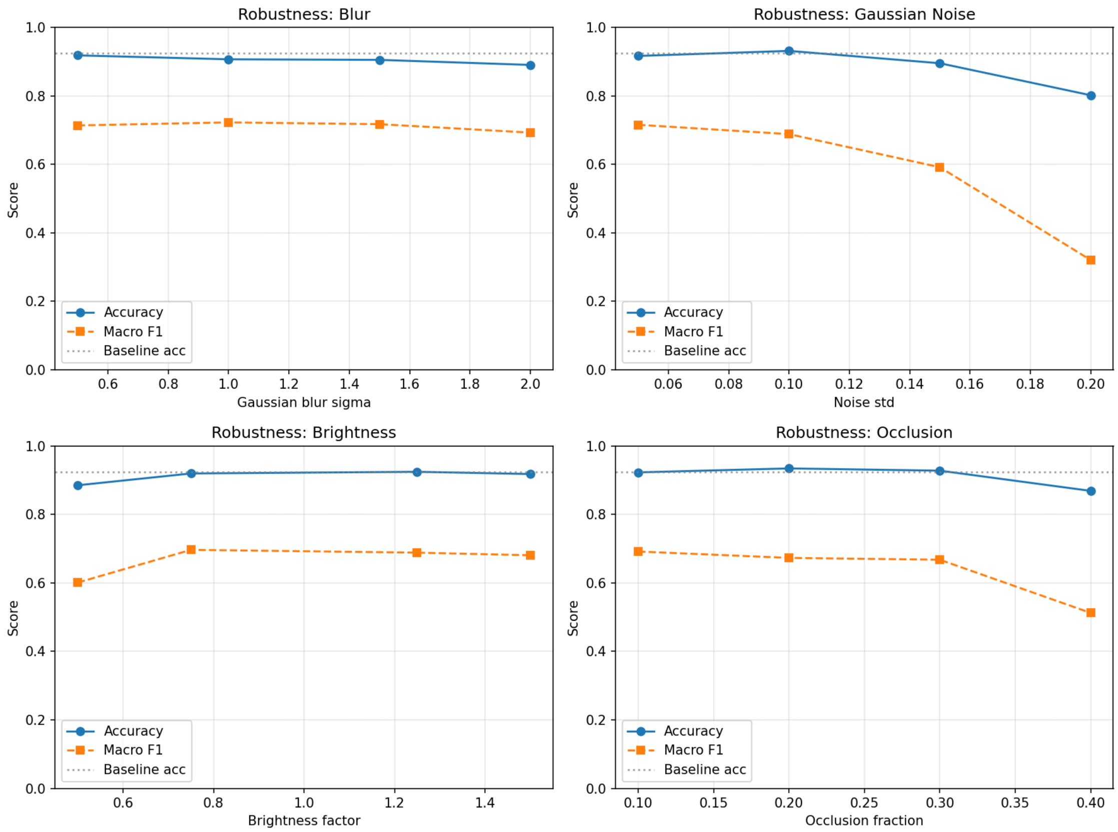


Figure 8: Accuracy and macro F1-score of the custom convolutional neural network under increasing severities of Gaussian blur, additive Gaussian noise, brightness scaling, and square occlusion. The dotted line marks the clean-data baseline accuracy. The model degrades gracefully under blur, brightness change, and moderate occlusion but is markedly sensitive to heavy additive noise, where the macro F1-score falls steeply.